Semiring-Based Linear Algebra Framework for Two-Way Regular Path Queries

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Abstract. Two-way regular path queries (2-RPQs) allow the use of regular languages over edges and their inverses in edge-labeled graphs to constrain paths of interest. Adopted in modern graph query languages such as GQL, SQL/PGQ, and Cypher, 2-RPQs have become part of the GQL ISO standard. However, their efficient evaluation on large-scale graphs remains challenging. In this paper, we propose a semiring-based generalization of the linear algebra approach for evaluating single-source 2-RPQs. The original LARPQ algorithm solves the reachability problem using Boolean matrices. We generalize it by replacing Boolean algebra with arbitrary semirings, enabling a unified solution for various path-aggregation problems. Specifically, we employ the path semiring to find all simple paths and all trails. The algorithms are implemented using the SuiteSparse:GraphBLAS library within the LAGraph infrastructure. Experimental evaluation on real-world knowledge graphs demonstrates significant performance improvements over state-of-the-art solutions.

Keywords: Regular path queries · Two-way RPQ · Linear algebra · Semirings · Graph databases · Sparse matrices.

1 Introduction

Language-constrained path querying (FLPQ) [5] provides a mechanism for searching paths in edge-labeled graphs where constraints are specified in terms of formal languages. A path is considered valid if the word formed by concatenating the labels of its edges belongs to the specified language. When constraints are regular languages, such queries are called regular path queries (RPQs). In practice, it is often useful to allow traversal opposite to edge directions, leading to two-way regular path queries (2-RPQs) [10, 7]. Queries specifying only a start or an end vertex are known as single-source or single-destination queries, respectively. Both have been adopted in graph query languages such as GQL [14], SQL/PGQ, and SPARQL [1].

Graph database systems support different path query semantics, as the choice significantly affects both result interpretation and computational complexity.

Common semantics include: reachability (all vertices reachable via any valid path), all-shortest-paths (one shortest path per reachable vertex), all-simple-paths (all simple paths satisfying the query), and all-trails (all trails satisfying the query) [13]. Each semantics serves different use cases, and the query engine must select the appropriate algorithm accordingly.

Efficient evaluation of RPQs and 2-RPQs remains an active research area [11, 20]. Sparse linear algebra-based methods have shown promise for RPQ evaluation [3]. In particular, the LARPQ (Linear Algebra-based Regular Path Query) algorithm [6] employs a breadth-first traversal that simultaneously processes both the graph and the finite automaton representing the regular language. While LARPQ demonstrates promising performance, it was designed solely for reachability, and its application to path-related semantics (e.g., simple or shortest paths) remains unexplored.

The contributions of this paper are as follows.

- A formal generalization of the LARPQ algorithm to a semiring-based framework, enabling the solution of various path-aggregation problems within a unified algorithmic structure.
- Instantiation of the framework with specific semirings to solve distinct problems: Boolean semiring for reachability, and different path semirings with index unary operators for all-simple-paths and all-trails and all-shortest-paths semantics.
- Experimental evaluation on both real-world and synthetic knowledge graphs and queries demonstrating significant performance improvements over state-of-the-art alternatives.

2 Preliminaries

In this section, we introduce the foundational concepts required throughout the paper, including edge-labeled graphs, paths, finite automata, and regular path queries.

2.1 Edge-Labeled Graphs

We begin by defining the graph data model used in this work.

Definition 1 (Edge-labeled graph). *A quadruple $G = \langle V, E, L, \lambda_G \rangle$ is called an edge-labeled graph (or simply a graph) if:*

- *V is a finite set of vertices;*
- *$E \subseteq V \times V$ is a finite set of edges;*
- *L is a finite set of labels;*
- *$\lambda_G : E \mapsto 2^L$ represents edge labels.*

For the remainder of this paper, we assume that vertices of any graph $G = \langle V, E, L, \lambda_G \rangle$ are enumerated, and without loss of generality, $V = \{1, 2, \ldots, |V|\}$.

To enable traversal in both directions along edges, we introduce the notion of inverted edges and labels. Let $G = \langle V, E, L, \lambda_G \rangle$ be an edge-labeled graph. We define:

- $a^- \notin L$ for each $a \in L$, with $(a^-)^- = a$ and $a = b \Leftrightarrow a^- = b^-$;
- $e^- = (v, u)$ where $e = (u, v)$;
- $E^- = \{e^- \mid e \in E\}$;
- $L^- = \{a^- \mid a \in L\}$;
- $\lambda_G^- : E^- \mapsto 2^{L^-}$, where $\lambda_G^-(e^-) = \{a^- \mid a \in \lambda_G(e)\}$.

We also define the bidirectional edge and label sets:

- $E^{\leftrightarrow} = E \cup E^-$;
- $L^{\leftrightarrow} = L \cup L^-$;
- $\lambda_G^{\leftrightarrow} : E^{\leftrightarrow} \mapsto 2^{L^{\leftrightarrow}}$, where $\lambda_G^{\leftrightarrow}(e) = \lambda_G(e) \cup \lambda_G^-(e)$;
- $G^{\leftrightarrow} = \langle V, E^{\leftrightarrow}, L^{\leftrightarrow}, \lambda_G^{\leftrightarrow} \rangle$.

The graph $G^{\leftrightarrow}$ allows traversal along both original and inverted edges, which is essential for evaluating two-way regular path queries.

2.2 Paths

Definition 2 (Path). *A path $\pi = (e_1, e_2, \ldots, e_n)$ of length n in the graph $G = \langle V, E, L, \lambda_G \rangle$ from u_1 to v_n is a finite sequence of edges $e_i = (u_i, v_i) \in E$ such that $\forall 1 \leq i \leq n-1 : v_i = u_{i+1}$.*

We denote $|\pi|$ as the length of a path π and consider zero-length paths, represented by an empty sequence from a vertex to itself.

Definition 3 (Simple path). *A path $\pi = (e_1, e_2, \ldots, e_n)$ is simple if it contains no repeated vertices. Formally, assuming $e_i = (v_i, u_i)$, this means that all vertices in the sequence $v_1, u_1, u_2, \ldots, u_n$ are pairwise distinct.*

Definition 4 (Trail). *A path $\pi = (e_1, e_2, \ldots, e_n)$ is a trail if it contains no repeated edges. Formally, $\forall 1 \leq i, j \leq n : e_i = e_j \Rightarrow i = j$.*

Definition 5 (Two-way path). *A two-way path (2-path) $\pi = (e_1, e_2, \ldots, e_n)$ in G is a path in $G^{\leftrightarrow}$. Two-way simple paths and two-way trails are defined analogously.*

The mapping ω associates paths with words over the label alphabet:

- $\omega_G(\pi) = \{a_1 \cdot \ldots \cdot a_n \mid a_i \in \lambda_G(e_i)\}$ for a path $\pi = (e_1, \ldots, e_n)$ in G;
- $\omega_{G^{\leftrightarrow}}(\pi) = \{a_1 \cdot \ldots \cdot a_n \mid a_i \in \lambda_G^{\leftrightarrow}(e_i)\}$ for a 2-path $\pi = (e_1, \ldots, e_n)$ in G.

Here, $\cdot$ denotes string concatenation.

2.3 Two-Way Finite Automata

Regular languages, recognized by finite automata, provide the foundation for specifying path constraints. For two-way path queries, we require a variant of nondeterministic finite automata.

Definition 6 (Two-way NFA). *A two-way nondeterministic finite automaton (2-NFA) is a tuple* $N = \langle Q, \Sigma, \Delta, \lambda_N, Q_S, Q_F \rangle$*, where:*

- Q *is a finite set of states;*
- Σ *is a finite alphabet;*
- $\Delta \subseteq Q \times Q$ *is the transition relation;*
- $\lambda_N : \Delta \mapsto 2^{\Sigma^{\leftrightarrow}}$ *maps transitions to labels;*
- $Q_S \subseteq Q$ *is the set of starting states;*
- $Q_F \subseteq Q$ *is the set of final states.*

Definition 7 (Two-way DFA). *A two-way deterministic finite automaton (2-DFA) is a special case of 2-NFA with* $|Q_S| = 1$ *and* $\forall (q, q'), (q, q'') \in \Delta : q' \neq q'' \Rightarrow \lambda_N((q, q')) \cap \lambda_N((q, q'')) = \varnothing$.

A 2-NFA can be viewed as a graph $G_N = \langle Q, \Delta, \Sigma^{\leftrightarrow}, \lambda_N \rangle$ equipped with sets of starting and final states. Conversely, a graph $G = \langle V, E, L, \lambda_G \rangle$ equipped with $V_S \subseteq V$ can be seen as a 2-NFA $N_G = \langle V, L, E, \lambda_G^{\leftrightarrow}, V_S, V \rangle$ that accepts:

$$[[N_G]] = \bigcup_{\text{2-path } \pi_G \text{ in } G \text{ from } v_S \in V_S} \omega_{G^{\leftrightarrow}}(\pi_G)$$

2.4 Two-Way Regular Path Queries

Definition 8 (Two-way regular path query). *A two-way regular path query (2-RPQ) is a triple* $\langle G, R, v_s \rangle$ *where* $G = \langle V, E, L, \lambda_G \rangle$ *is a graph,* R *is a regular language over the alphabet* $L^{\leftrightarrow}$*, and* $v_s \in V$ *is a starting vertex.*

We define a *simple path* as a path without repeated vertices and a *trail* as a path without repeated edges. Path length is measured as the number of edges. In this setting, we define the following semantic variants of 2-RPQs considered in this paper.

- **S**ingle-**s**ource **s**hortest **p**aths (**SSSP**):

$$[[Q]]_{SSSP} = \{\pi \mid \pi \text{ is a shortest path to any } v \text{ in } G \text{ from } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

- **S**ingle-**d**estination **s**hortest **p**aths (**SDSP**):

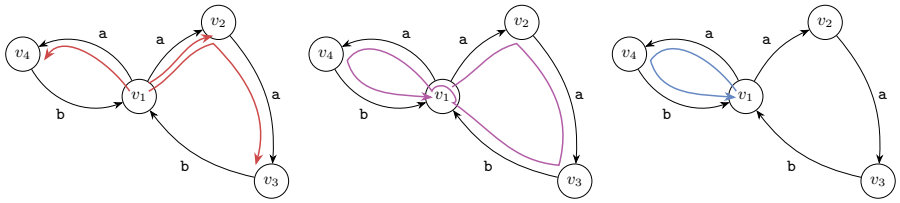
$$[[Q]]_{SDSP} = \{\pi \mid \pi \text{ is a shortest path from any } v \text{ in } G \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

- **S**ingle-**s**ource **a**ll **s**imple **p**aths (**SSASP**):

$$[[Q]]_{SSASP} = \{\pi \mid \pi \text{ is a 2-simple path in } G \text{ from } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

- **S**ingle-**d**estination **a**ll **s**imple **p**aths (**SDASP**):

$$[[Q]]_{SDASP} = \{\pi \mid \pi \text{ is a 2-simple path in } G \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$



(a) All simple paths (b) One of the longest 2-paths in all trails (c) All shortest paths from v_1 to v_1

Fig. 1: Different query semantics for $((a^+)\,(b?))^+$

– **S**ingle-**s**ource **a**ll **t**rails (**SSAT**):

$$[[Q]]_{SSAT} = \{\pi \mid \pi \text{ is a 2-trail in } G \text{ from } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

– **S**ingle-**d**estination **a**ll **t**rails (**SDAT**):

$$[[Q]]_{SDAT} = \{\pi \mid \pi \text{ is a 2-trail in } G \text{ to } v_s \text{ and } \omega_{G^{\leftrightarrow}}(\pi) \cap R \neq \varnothing\}.$$

Figure 1 presents examples of single-source query results under different semantics on the same graph: **a)** *all simple paths*, **b)** *all trails*, and **c)** *all shortest paths*.

3 Linear Algebra, Graphs and Relations

In this section, we establish the linear algebra foundations required for our approach. We introduce the concept of semirings, discuss matrix representations of relations, and present the original LARPQ algorithm for solving the reachability problem.

3.1 Semirings

Semirings provide an algebraic framework that generalizes both Boolean algebra and path-based computations. A semiring consists of a set equipped with two binary operations that satisfy certain axioms. In graph processing algorithms, with respect to GraphBLAS [15], semirings are defined differently from the ones in algebra. The domains of $\otimes$ can vary; distributivity and a multiplicative identity elements are optional.

Definition 9 (Semiring). *A semiring is a tuple* $\langle D_{out}, D_{in_1}, D_{in_2}, \oplus, \otimes, \mathbf{0}\rangle$ *where:*

– D_{out}, D_{in_1}, *and* D_{in_2} *are three domains*[1]*;*
– $\langle D_{out}, \oplus, \mathbf{0}\rangle$ *is a commutative monoid with an identity element* $\mathbf{0} \in D_{out}$*;*
– $\otimes : (D_{in_1} \times D_{in_2}) \to D_{out}$ *is a binary operator;*

Given a semiring $\mathcal{S} = \langle D_{out}, D_{in_1}, D_{in_2}, \oplus, \otimes, \mathbf{0}\rangle$ we define *a dual semiring* $\overline{\mathcal{S}} = \langle D_{out}, D_{in_2}, D_{in_1}, \oplus, \overline{\otimes}, \mathbf{0}\rangle$ having $\overline{\otimes}(a, b) = \otimes(b, a)$.

[1] Actually, types in respective programming language.

3.2 Matrix Representations

Relations between finite sets can be represented using matrices. For two enumerated finite sets $A = \{1, 2, \ldots, n\}$ and $B = \{1, 2, \ldots, m\}$, a binary matrix T of size $n \times m$ can represent a relation $T \subseteq A \times B$ where $T_{ij} = 1 \Leftrightarrow (i, j) \in T$.

Let $G = \langle V, E, L, \lambda_G \rangle$ be an arbitrary graph and $N = \langle Q, \Sigma, \Delta, \lambda_N, Q_S, Q_F \rangle$ an arbitrary 2-NFA. For each label $a \in L^{\leftrightarrow}$, we define:

- $E_a = \{e \mid e \in E, a \in \lambda_G^{\leftrightarrow}(e)\}$;
- $\Delta_a = \{\delta \mid \delta \in \Delta, a \in \lambda_N(\delta)\}$.

Definition 10 (Adjacency matrix of a label). *Let G_a be the matrix representing the relation E_a. Then G_a is called the adjacency matrix of label a.*

Definition 11 (Boolean decomposition). *A Boolean decomposition of the adjacency matrix G is the set of Boolean matrices $\{G_a \mid a \in L^{\leftrightarrow}\}$.*

3.3 The Path Semiring

For solving path enumeration problems (i.e. to search all simple paths, all trails), we require a more expressive semiring based on path representations.

Let $\mathrm{D}_{\textsc{Paths}} = 2^{\mathbb{N}^*}$, where $\pi' = (v_1, v_2, \ldots, v_n) \in \mathbb{N}^*$ represents a 2-path $\pi = ((v_1, v_2), (v_2, v_3), \ldots, (v_{n-1}, v_n))$ of length $n - 1$. For $a, b \in \mathrm{D}_{\textsc{Paths}}$ and $c \in \mathbb{B}$, we define the paths semiring and its dual:

- $\textsc{Paths} = \langle \mathrm{D}_{\textsc{Paths}}, \mathrm{D}_{\textsc{Paths}}, \mathbb{B} = \{0, 1\}, \cup, \texttt{fst}, \varnothing \rangle$
  with $\texttt{fst}(x, y) = \begin{cases} x & \text{if } y = 1 \\ \varnothing & \text{otherwise} \end{cases}$;
- $\overline{\textsc{Paths}} = \langle \mathrm{D}_{\textsc{Paths}}, \mathbb{B} = \{0, 1\}, \mathrm{D}_{\textsc{Paths}}, \cup, \texttt{snd}, \varnothing \rangle$
  with $\texttt{snd}(x, y) = \begin{cases} y & \text{if } x = 1 \\ \varnothing & \text{otherwise} \end{cases}$.

Let $A = (a_{ij})$, $B = (b_{ij})$ be matrices over $\mathrm{D}_{\textsc{Paths}}$ and $C = (c_{ij})$ be a Boolean matrix. We extend the operations to matrices:

- **Sum:** $A_{n \times m} \oplus B_{n \times m} = (a_{ij} \oplus b_{ij})_{n \times m}$;
- **Product:** $A_{n \times m} \otimes C_{m \times k} = (\bigoplus_{l=1}^{m} a_{il} \otimes c_{lj})_{n \times k}$;
- **Zero-matrix:** $\mathbf{0}_{n \times k}^{\textsc{Paths}} = (\varnothing)_{n \times k}$.

3.4 Index Unary Operators

The path semiring alone is insufficient for enforcing path semantics (simple paths, trails). We introduce index unary operators that can filter paths based on vertex or edge repetition.

Definition 12 (Index unary operator). *Let $A_{n \times m} = (a_{ij})$ be a matrix over a semiring R. An index unary operator $I : R \times \mathbb{N} \times \mathbb{N} \mapsto R$ is applied element-wise: $I(A) = (I(a_{ij}, i, j))_{n \times m}$.*

Given a Boolean matrix P of size $n \times m$ we define a specific index unary operators for $n \times m$ matrices over $\mathbb{B} = \{0, 1\}$,

- $I^P_{Reach}(a, i, j) = \begin{cases} 1 & \text{if } a = 1 \text{ and } P_{ij} = 0 \\ 0 & \text{otherwise.} \end{cases}$

Given a $\mathrm{D}_{\mathrm{PATHS}}$ matrix P of size $n \times m$ for $n \times m$ matrices over $\mathbb{B} = \{0, 1\}$ we define:

- $I_{Simple}(a, i, j) = \left\{ (v_1, \ldots, v_n, j) \,\middle|\, \begin{array}{l} (v_1, \ldots, v_n) \in a \\ \forall t : 1 \leq t \leq n : v_t \neq j \end{array} \right\};$
- $I_{Trail}(a, i, j) = \left\{ (v_1, \ldots, v_n, j) \,\middle|\, \begin{array}{l} (v_1, \ldots, v_n) \in a \\ \forall t : 1 \leq t \leq n-1 : (v_t, v_{t+1}) \neq (v_n, j) \end{array} \right\}.$
- $I^P_{Shortest}(a, i, j) = \begin{cases} \{(v_1, \ldots, v_n, j) \mid (v_1, \ldots, v_n) \in a\} & \text{if } \forall k \, P_{kj} = \varnothing \\ \varnothing & \text{otherwise.} \end{cases}$

GB

ALGIN TO EQUATION

The operator I^P_{Reach} behaves like a binary mask when applied to a matrix, enforcing the exclusion of previously visited states. The operator I_{Simple} extends paths while filtering out those that would introduce repeated vertices, consequently enforcing simple paths. The operator I_{Trail} filters out extensions that would introduce repeated edges, thus enforcing trails. The operator $I^P_{Shortest}$ combines path extension with the logic of I^P_{Reach}: when extending a path, it checks whether the newly introduced vertex already appears in the matrix P, as a result, only shortest paths candidates remain.

4 A Semiring Framework for Single-Source RPQs

In this section, we present our main contribution: a generalized semiring-based framework for evaluating single-source 2-RPQs. The original LARPQ algorithm operates over the Boolean semiring to solve reachability. We show that by replacing the Boolean semiring with other algebraic structures, the same algorithmic framework can solve different path-aggregation problems.

4.1 Generalized Algorithm

The key insight is that the LARPQ algorithm can be abstracted to work over any semiring equipped with appropriate operations. Algorithm 2 presents the generalized version.

The generalized algorithm differs from the original in several key aspects:

- Matrix operations ($\oplus$, $\otimes$) are performed over an arbitrary semiring $\mathcal{S}$ having $D_{in_2} = \mathbb{B}$;
- An optional index unary operator I^P can be applied to filter or transform path information;
- The final result format depends on the semiring instantiation.

We now present two important instantiations of this framework.

Input: Semiring $\mathcal{S} = \langle D_{out}, D_{in_1}, \mathbb{B}, \oplus, \otimes, \mathbf{0} \rangle$, index unary operator I^P, 2-NFA $\mathcal{N} = \langle Q, \Sigma, \Delta, \lambda_{\mathcal{N}}, Q_S, Q_F \rangle$, graph $\mathcal{G} = \langle V, E, L, \lambda_{\mathcal{G}} \rangle$, source $v_s \in V$
Output: Result depending on semiring instantiation

1. $\{N^a\} \leftarrow$ Boolean decomposition of the $\mathcal{N}$ adjacency matrix
2. $\{G^a\} \leftarrow$ Boolean decomposition of the $\mathcal{G}$ adjacency matrix
3. $M \leftarrow \mathbf{0}$, $P \leftarrow \mathbf{0}$
4. For each $q \in Q_S$: $M_{qv_s} \leftarrow \mathbf{1}$
5. While $M \neq \mathbf{0}$:
   - If I^P defined: $M \leftarrow I(M)$
   - $P \leftarrow P \oplus M$
   - $M \leftarrow \bigoplus_{a \in \Sigma^{\leftrightarrow} \cap L^{\leftrightarrow}} ((N^a)^T \overline{\otimes} M \otimes G^a)$
6. $F_{1 \times |Q|} \leftarrow \mathbf{0}_{1 \times |Q|}$
7. For each $q \in Q_F$: set $F_{1q} \leftarrow 1$
8. Return $P^F = F \overline{\otimes} P$

Fig. 2: Generalized Single-Source 2-RPQ using Semirings

4.2 Instantiation I: Boolean Semiring for Reachability

By instantiating $\mathcal{S}$ with the Boolean semiring $\langle \mathbb{B} = \{0, 1\}, \mathbb{B}, \mathbb{B}, \vee, \wedge, 0 \rangle$ and the index unary operator I^P with I^P_{Reach}, we recover the original LARPQ algorithm for reachability from [6]. The matrix entries are Boolean values, where 1 indicates reachability. The index unary operator ensures each pair of an automaton state and vertex is processed only once.

The result is a Boolean vector P^F where $P^F_v = 1$ if and only if vertex v is reachable from the source via a path satisfying the regular language constraint.

4.3 Instantiation II: Path Semiring with Index Operators for All Paths

For enumerating all simple paths or all trails, we use the previously introduced path semiring $\langle \mathrm{D}_{\textsc{Paths}}, \mathrm{D}_{\textsc{Paths}}, \mathbb{B} = \{0, 1\}, \cup, \mathbf{fst}, \varnothing \rangle$ combined with appropriate index unary operators.

When $I^P = I_{Simple}$, the algorithm enumerates all simple 2-paths satisfying the query. When $I^P = I_{Trail}$, it enumerates all trails. The key difference from the reachability algorithm is that the frontier matrix M now stores actual path information (as sequences of vertices) rather than Boolean values. The index unary operator filters extensions that would violate the path semantics:

- I_{Simple} prevents extending a path with a vertex already present in the path (except the starting vertex), ensuring simplicity;
- I_{Trail} prevents extending a path with an edge already traversed, ensuring the result is a trail.
- I^P_{Simple} prevents extending a path with a vertex if shorter paths to that vertex were already accessed.

The algorithm maintains the BFS structure: on iteration n, M_{qv} contains all valid paths of length n from v_s to v that correspond to paths from some $q_s \in Q_S$ to q in the automaton. Figure 3 illustrates an example of the algorithm step for the all simple paths semantics.

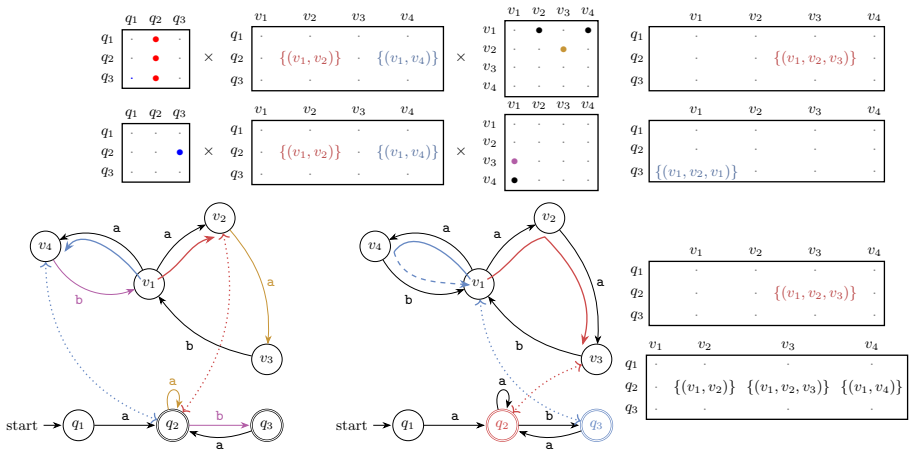


Fig. 3: The algorithm step for graph and automaton for regular expression $((a^+)(b?))^+$ using path semiring

The result is a Boolean vector P^F where P^F_v contains all 2-paths (all simple paths, all trails, all shortest paths) from v_s to v satisfying the regular language constraint.

4.4 Correctness Argument

The correctness argument follows the same induction scheme as in our previous work on LARPQ [6, Appendix A]. The induction is over the iterations of the main loop: the frontier matrix M stores exactly the paths produced at the current traversal depth, and the accumulator matrix P stores the union of all paths produced so far. Each iteration extends paths consistently in the query automaton and in the input graph; the optional index unary operator filters out extensions that violate the selected path semantics. After termination, selecting final automaton states yields exactly the query result for the chosen semiring instantiation.

5 Implementation Details

We have implemented the proposed algorithms within the LAGraph project [18], a collection of linear algebra-based graph processing algorithms built on top of

SuiteSparse:GraphBLAS [12]. This section describes the key implementation aspects.

The main engineering limitation is that SuiteSparse:GraphBLAS user-defined semiring values must have a fixed size to work with built-in automatic allocators. At the same time, an element of the path semiring may contain paths of different lengths and may accumulate an unbounded number of paths during evaluation. To fit these variable-size objects into the fixed-size semiring interface, the implementation uses a compact in-place representation for common cases and falls back to explicit memory management only when this representation overflows.

We introduce two constants, L_{fast} and C_{fast}, to control the maximum path length and the maximum cardinality of a $\mathrm{D}_{\mathrm{PATHS}}$ element that can be stored without external memory management. As long as all paths in an element are no longer than L_{fast} and the element contains at most C_{fast} paths, the value is stored directly inside the fixed-size semiring object. This fast representation avoids extra allocations and is intended to cover the typical case.

If a path $\pi \in M_{qv}$ with $|\pi| > L_{fast}$ appears, a new linked list representing this long path is allocated in a supplementary memory arena. Similarly, if $|M_{qv}| > C_{fast}$ for some q, v, an extensible linked list is allocated. These fallback structures keep the semiring value itself fixed-size: it stores only the inline data or references to the arena-managed objects. We used $L_{fast} = 8$ and $C_{fast} = 1$, empirically chosen to achieve the best performance in evaluation.

These two values provide a trade-off between peak memory consumption and execution speed. In practice, these constants should be adjusted for each dataset individually.

Finally, we ship the semiring-based algorithm implementation as a standalone executable file. A preprocessed graph, decomposed into a set of Boolean Matrix Market matrices, and a set of regular path query patterns described in a SPARQL-like query language are consumed as input and executed one by one. The CLI flag `-semantics` can be used to switch between all-simple-path, all-shortest-path and all-trail semantics.

6 Evaluation

We evaluate the proposed solution (LARPQ) on three semantics (all-simple-paths, all-shortest-paths, and all-trails) using real-world knowledge graphs. The experiments are designed to answer the following research questions.

- **RQ1:** How does our implementation compare with production graph database systems?
- **RQ2:** What is the performance behavior across different query types and graph structures?

Experiments are conducted on a workstation equipped with a Ryzen 9 7900X (4.7 GHz, 12 cores), 128 GB of DDR5 RAM, running Ubuntu 22.04. The timeout is set to 60 seconds per query. Each query is executed five times: one warm-up run followed by four measurement runs.

Competitors We compare the performance of the proposed semiring-based LARPQ algorithm against state-of-the-art graph database management systems.

- **MillenniumDB** or MDB (RPQ challenge version) [21]: a persistent, open-source graph database with state-of-the-art RPQ performance and natural support of all three path semantics.
- **FalkorDB** (v4.18.5, formerly RedisGraph [9]): an open-source in-memory property-graph database using GraphBLAS. Among the three path semantics, only all-trails is naturally expressed in FalkorDB, so we measure only the corresponding queries.

For all competitors, we use their respective query languages to specify queries. The time required for query processing (including parsing and optimization) is included in the measurements. All data is preloaded into storage, so graph preprocessing time is excluded from the results.

Dataset We used three different graphs and their respective sets of queries.

Wikidata [19]: 610 million edges, 91 million vertices, and 1 400 distinct labels. A total of 578 queries were filtered from a dump of 660 queries.

Yago-2S: 46 million edges, 7 million vertices, and 42 distinct labels. Seven complex queries were taken from [2].

RPQBench [22]: a synthetic dataset with 150 million edges, 57 million vertices, and 9 distinct labels. It includes 20 different query types, 10 queries were generated for each.

6.1 Experimental Results

We measure the total time for all queries, the mean and median time over all queries, and the mean and median per-query speedup over the proposed solution. A summary for all three graphs is presented in Tables 1, 3, and 2. Per-query results for Wikidata are shown in Figure 4. Only queries that were successfully evaluated for all competitors are included in these statistics. Figure 4d additionally shows queries completed by MillenniumDB and our solution but not by FalkorDB.

Table 1: Performance Summary on Wikidata (time in milliseconds)

Metric	Simple		Shortest		Trails		
	LARPQ	**MDB**	**LARPQ**	**MDB**	**LARPQ**	**MDB**	**FalkorDB**
Total time	**224 682**	1 111 429	**205 629**	780 011	152 128	921 506	**81 519**
Mean time	**428**	2 121	**392**	1 488	312	1 892	**167**
Median time	**1.7**	36.6	**2.0**	27.9	**1.4**	36.9	4.5
Mean Speedup	**1.0**	0.6	**1.0**	0.8	**1.0**	0.5	0.9
Median Speedup	**1.0**	0.1	**1.0**	0.1	**1.0**	0.1	0.7

Table 2: Performance Summary on RPQBench (time in milliseconds)

Metric	Simple		Shortest		Trails		
	LARPQ	**MDB**	**LARPQ**	**MDB**	**LARPQ**	**MDB**	**FalkorDB**
Total time	**857**	2836	**890**	3 253	**849**	3 256	1 685
Mean time	**43**	142	**45**	162	**53**	203	105
Median time	0.3	**0.2**	0.3	**0.1**	0.3	**0.2**	**0.2**
Mean Speedup	1.0	**1.5**	1.0	**2.1**	1.0	**1.5**	1.2
Median Speedup	1.0	**1.8**	1.0	**2.4**	1.0	**1.7**	1.5

Table 3: Performance Summary on YAGO-2S (time in milliseconds)

Metric	Simple[1]		Shortest		Trails[1]		
	LARPQ	**MDB**	**LARPQ**	**MDB**	**LARPQ**	**MDB**	**FalkorDB**[2]
Total time	**105**	215	**1 195**	1 956	**66**	479	—
Mean time	—	—	**171**	279	—	—	—
Median time	—	—	**167**	268	—	—	—
Mean Speedup	—	—	**1.0**	0.5	—	—	—
Median Speedup	—	—	**1.0**	0.6	—	—	—

[1] Only two queries of seven executed successfully by all competitors.
[2] Timeout for all queries.

The evaluation shows that execution time strongly depends on the particular query and the choice of start and target vertices. We define a query as *simple* for a system if it requires relatively little time to evaluate, and *hard* if, conversely, it requires a relatively large amount of time.

For RPQBench (Table 2), the total and mean execution times of our solution are more than three times better than those of the competitors across all semantics. However, the median time and per-query speedup favor the competitors. This indicates that the dataset contains a relatively large number of simple queries, while our solution exhibits a significant advantage on harder queries.

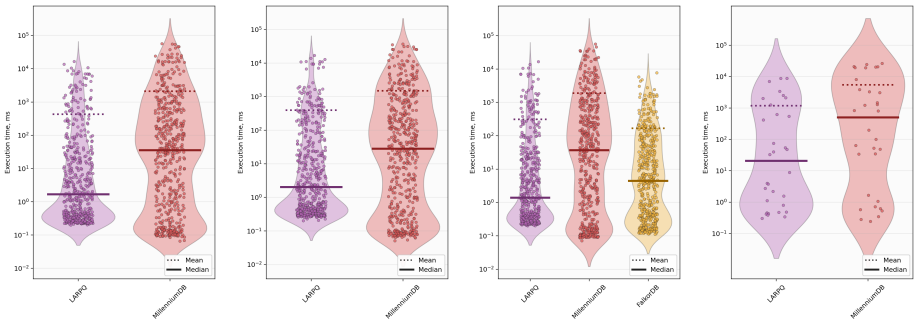
This observation is consistent with the results presented in Figure 4, where MillenniumDB outperforms our solution on simple queries but is slower on hard ones. The tail of queries that are challenging for all solutions is thinner for LARPQ than for MillenniumDB and FalkorDB, as illustrated in Figure 4c. Additionally, we observe that in the common-trails setting, FalkorDB achieves the best total and mean times, while LARPQ achieves the best median time, which is also consistent with the shape of the hard queries tail.

Analysis of the YAGO-2S dataset (see table 3) shows that the shortest-path semantics is the only one for which all seven queries completed successfully across all competing systems. For the simple-paths and trails semantics, many queries timed out by all solutions. Nevertheless, for queries that completed successfully, our solution demonstrates better performance than the competitors.

A detailed analysis of which queries are *simple* and which are *hard* for our solution yields results similar to those reported in [6], indicating that the proposed generalization significantly inherits the properties of the basic version.

Specifically, LARPQ outperforms competitors on queries with complex patterns over labels corresponding to many edges (e.g., $l_1 \cdot l_2 \cdot (l_3 \mid l_4)^*$), while on queries with simple patterns such as $l_1^* \cdot l_2$, it shows no performance advantage, and MillenniumDB may be a better choice.

Overall, the results indicate that LARPQ is a strong choice for complex RPQs, especially when query patterns involve labels with many matching edges, whereas competing systems may perform better on simpler patterns.



(a) Simple paths (b) Shortest paths (c) Trails (common for all competitors) (d) Trails (FalkorDB failed)

Fig. 4: Performance on Wikidata

7 Related Work

The algebraic approach to path problems has a long history. Kleene's algorithm computes reachability using matrix operations over the Boolean semiring. This approach has been generalized to arbitrary semirings for solving various algebraic path problems [4], including path enumeration. Different query semantics and respective algorithms for regular path queries, including all shortest paths and all simple paths, have been studied in the literature [16, 17, 20].

The GraphBLAS initiative [15, 8, 12] established a standardized interface for expressing graph algorithms using linear algebra operations. The GraphBLAS specification supports arbitrary semirings, enabling the expression of algorithms that go beyond traditional numeric computations. LAGraph [18] provides a collection of graph algorithms built on GraphBLAS, including BFS, PageRank, and triangle counting. Recent work has explored linear algebra approaches for various graph problems. However, RPQ evaluation in this paradigm remains underexplored, with only a few prior works [3, 6] addressing this problem.

8 Conclusion and Future Work

We have presented a semiring-based generalization of the linear algebra approach for evaluating two-way regular path queries. Source code and additional materials are available at GitHub[2]. Experimental evaluation demonstrates that the proposed approach achieves competitive performance with state-of-the-art graph database systems while offering a more flexible framework for different path semantics. At the same time, our evaluation reveals limitations of both the proposed and existing solutions in terms of performance (see the discussion of results on YAGO-2S and table 3), opening the door for further research on performance improvements.

Future work includes the use of additional semirings, such as counting semirings for path counting, as well as integration with property graphs and more expressive query languages such as CRPQs. Another direction is distributed implementation and acceleration through offloading matrix operations to GPUs.

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